$$0.3667\,\varphi_2 - 0.2\,\varphi_3 - 0.1\,\varphi_4 = 2.6667$$
$$-0.2\,\varphi_2 + 0.4778\,\varphi_3 - 0.1667\,\varphi_4 = -18.2222$$
$$-0.1\,\varphi_2 - 0.1667\,\varphi_3 + 0.3167\,\varphi_4 = 16.6667$$

Где

$$G_3 = \frac{1}{R_3} = \frac{1}{15} = 0.066667\ S$$

$$G_2 = \frac{1}{R_2} = \frac{1}{9} = 0.111111\ S$$

$$G_5 = \frac{1}{R_5} = \frac{1}{20} = 0.05$$

$$G_1 = \frac{1}{R_1} = \frac{1}{6} = 0.166667\ S$$

$$G_6 = \frac{1}{R_6} = \frac{1}{10} = 0.1\ S$$

Решаем систему с помощью матриц

$$\varphi = (A G_b A^T)^{-1} (A J_b - A G_b E_b)$$

После расчета получили

$$\varphi_2 = 10.557\ V$$

$$\varphi_3 = -17.394\ V$$

$$\varphi_4 = 46.802\ V$$

Определяем токи ветвей

$$I_3 = \frac{\varphi_1 - \varphi_2 + E_3}{R_3} = \frac{0 - (10.557) + 100}{15} = 5.963\ A$$

$$I_2 = \frac{\varphi_3 - \varphi_1 + E_2}{R_2} = \frac{-17.394 - (0) + 50}{9} = 3.623\ A$$

$$I_5 = \frac{\varphi_4 - \varphi_1}{R_5} = \frac{46.802 - (0)}{20} = 2.34\ A$$

$$I_1 = \frac{\varphi_3 - \varphi_4 + E_1}{R_1} = \frac{-17.394 - (46.802) + 100}{6} = 5.966\ A$$

$$I_6 = \frac{\varphi_4 - \varphi_2}{R_6} = \frac{46.802 - (10.557)}{10} = 3.626\ A$$

$$I_0 = (\varphi_3 - \varphi_2) G_0 = (-17.394 - (10.557))\ 0.2 = -5.589\ A$$

$$I_k = J_k = 4\ A$$